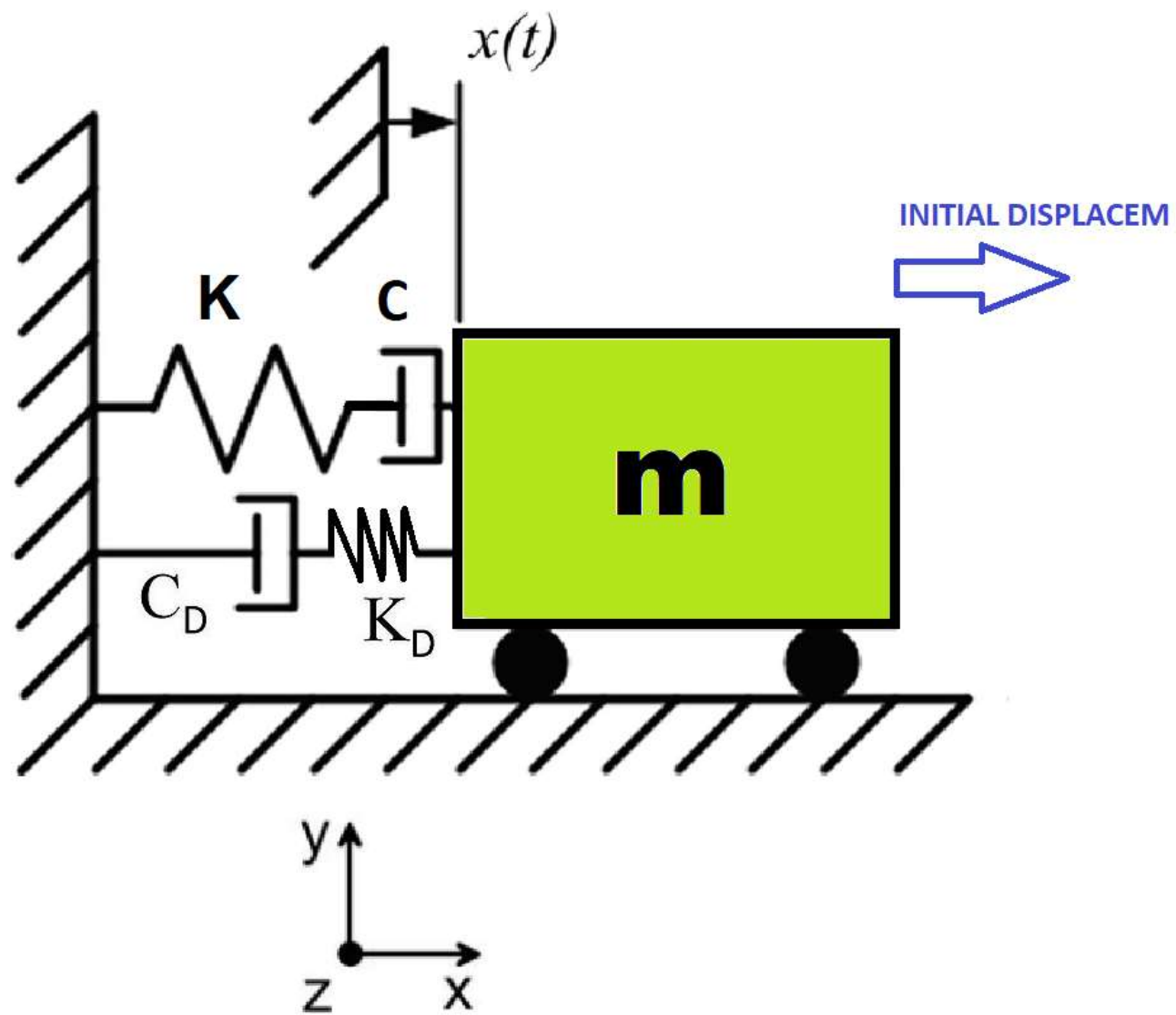


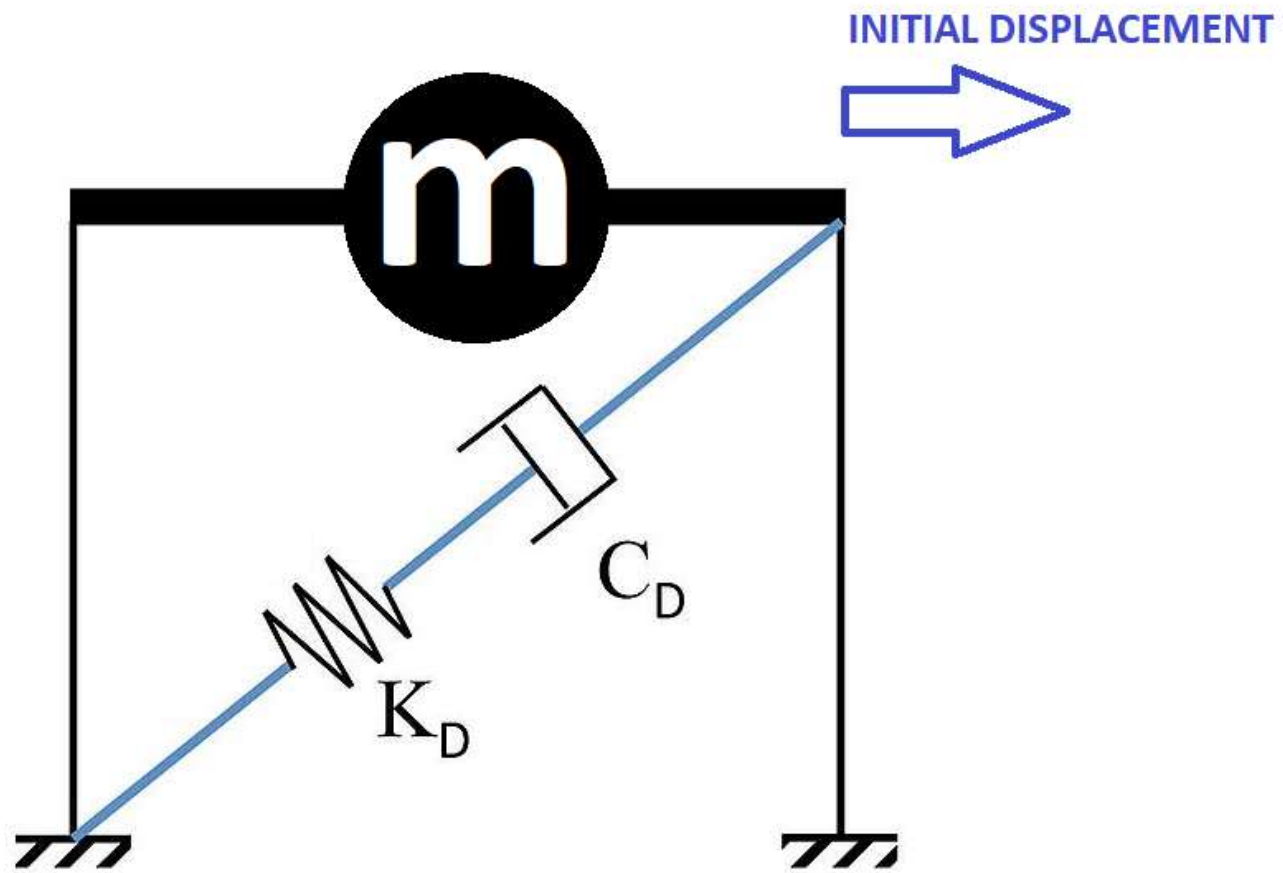
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

SENSITIVITY ANALYSIS BY CHANGING STRUCTURAL DAMPER STIFFNESS AND DAMPING RATIO USING OPENSEES

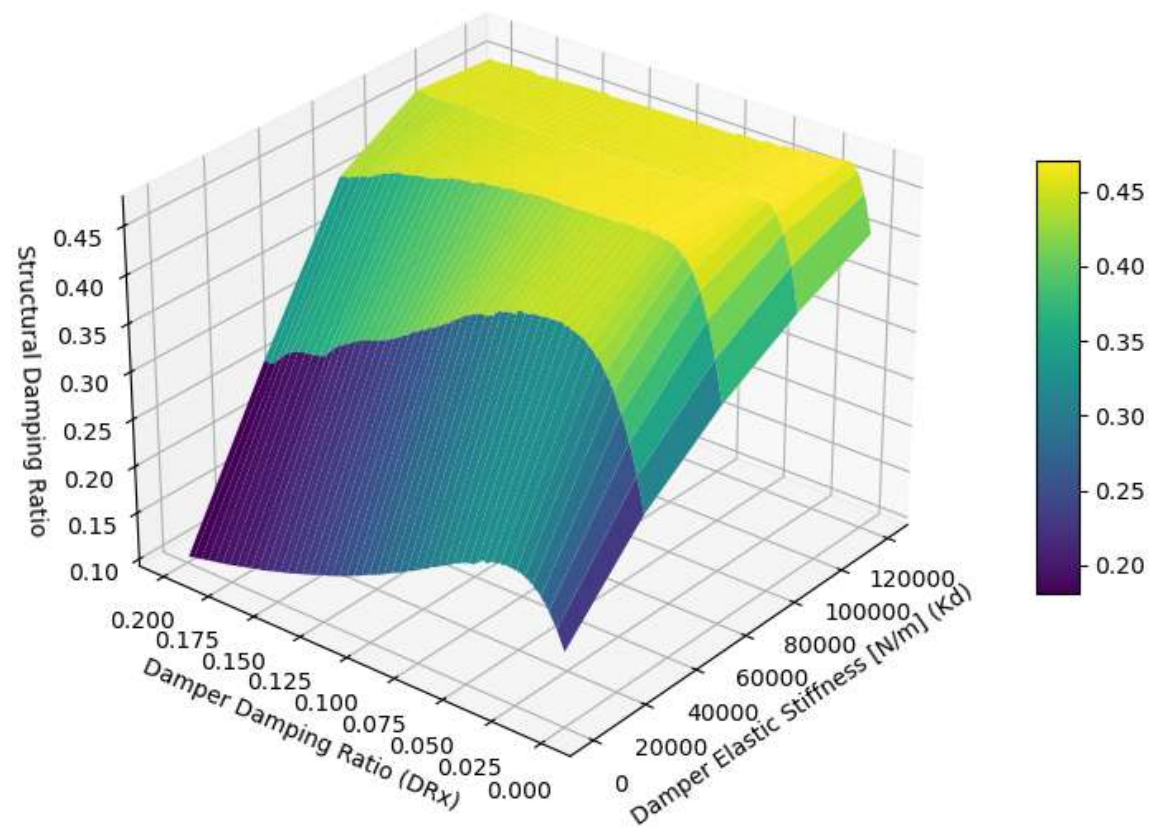
THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

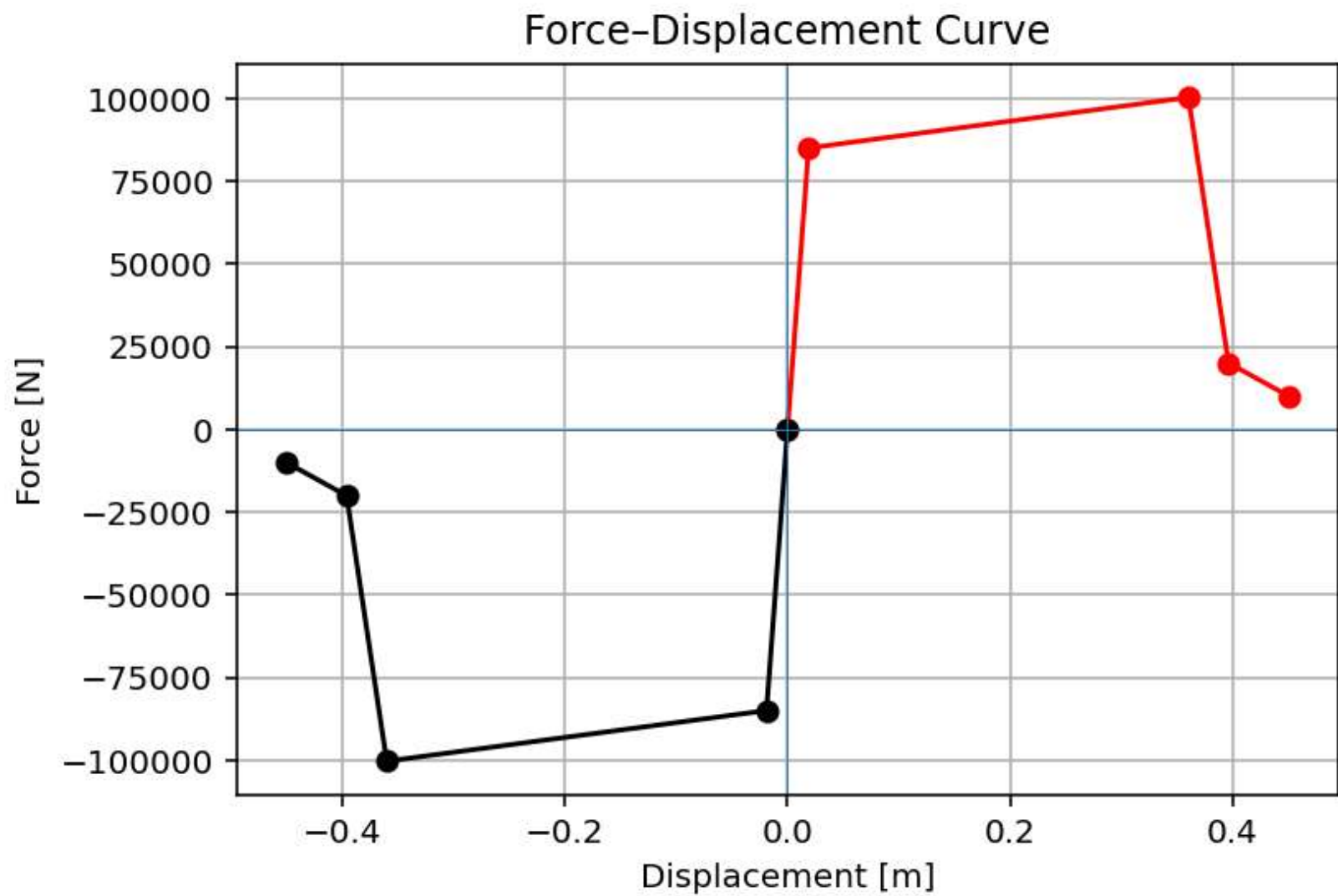
Suppose we have an existing steel or concrete structure and wish to enhance its seismic performance by adding supplemental dampers. The goal is to increase the overall effective damping of the structural system through the contribution of these devices. In this type of study a sensitivity analysis is carried out by varying key damper parameters; such as the elastic stiffness of the damper and its viscous damping coefficient in order to evaluate the resulting total damping of the structure. This total damping combines the inherent damping of the original structure and the additional damping provided by the dampers.





Damping Ratio vs Kd and DRx





Spyder (Python 3.12)

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SENSITIVITY_DRX_KD_Damper_SDOF.py

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MER
3 # SENSITIVITY ANALYSIS BY CHANGING STRUCTURAL DAMPER STIFFNESS AND DAMPI
4 # -----
5 # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI
6 # EMAIL: salar.d.ghashghaei@gmail.com
7 #####
8
9 """
10 Suppose we have an existing steel or concrete structure and wish to
11 enhance its seismic performance by adding supplemental dampers.
12 The goal is to increase the overall effective damping of the structural
13 system through the contribution of these devices. In this type of study
14 a sensitivity analysis is carried out by varying key damper parameters;
15 such as the elastic stiffness of the damper and its viscous damping
16 coefficient—in order to evaluate the resulting total damping of the
17 structure. This total damping combines the inherent damping of the
18 original structure and the additional damping provided by the dampers.
19 -----
20 Nonlinear Seismic Performance Assessment of a SDOF:
21 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static
22
23 This OpenSeesPy script performs rigorous nonlinear static and dynamic analys
24 for performance-based earthquake engineering. The 2D model incorporates bot
25 (elastic-perfectly plastic or hysteretic steel with strain hardening)
26 and geometric nonlinearity (corotational truss formulation) to capture P-de
27 and large displacements—essential for collapse assessment.
28
29 Eight analysis protocols are implemented:
30 (1) [PERIOD] : Structural Period
31 (2) [STATIC] : Gravity load analysis establishing dead load state
32 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity c
33 and plastic mechanism identification
34 (4) [CYCLIC DISPLACEMENT] : Symmetric cyclic displacement protocol capturing
```

Damping Ratio vs Kd and DRx

Structural Damping Ratio

Damper Elastic Stiffness [N/m] (K_d)

Damper Damping Ratio (DR_x)

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Qt Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 808, Col 59 UTF-8 CRLF RW Mem 69%